

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Minimum absorbency at 500 (nm) (1) Two maxima at 450 and 700 (nm) (1) Increases between 500 and 700 nm (1) Allow 'peaks, dips and peaks again' for (1)		3		3		
		(ii)		Selection of 450 (1) Manipulation $\frac{300\,000\,000}{450\,(\text{ecf}) \times 10^{-9}}$ (1) Answer = $6.667 / 6.7 \times 10^{14}$ [Hz] (1)	1	2		3	2	
	(b)	(i)		60% = 30 so 20% = 10 (1) therefore 40% = 20 (1)		2		2	2	
		(ii)		Plants are (generally) stationary / animals move more(1) so no energy / energy required for movement (1) OR foxes move more than / shrews / more than grasshoppers OR foxes have more body mass {than shrews / than grasshoppers} (1)	2			2		
		(iii)		Foxes move carrying seeds (1) Seeds / plant becomes more widely distributed (1)	2			2		
				Question 4 total	5	7	0	12	4	0